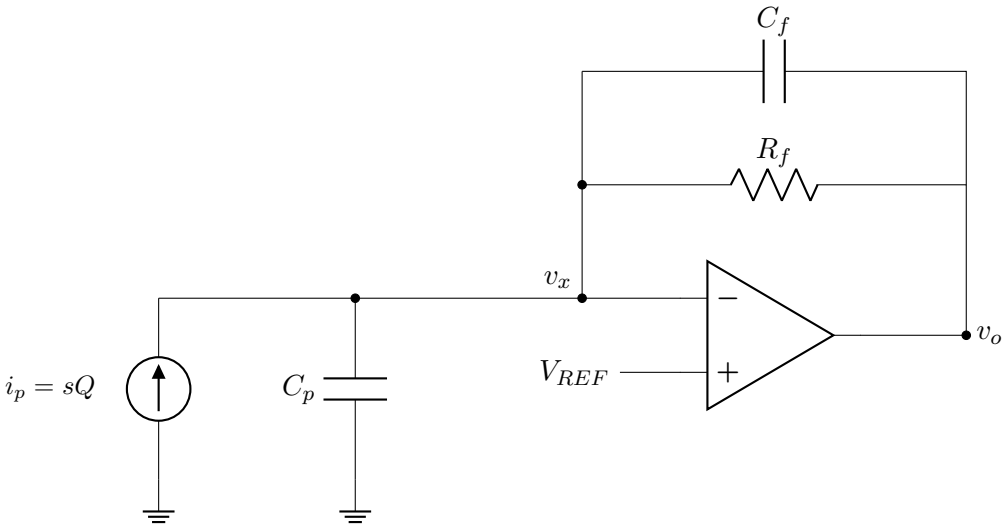
The sensor head, from the equations up

Charge amplifier, OPA365, DC point and noise

ADS7057 test board v3 — study notes

All voltages are *deviations from the DC operating point*, written in lowercase (v). Capital V is a total or DC value. The Laplace variable is $s = j\omega = j2\pi f$.

1 The circuit and its three equations



The piezo is a charge source Q in parallel with its own capacitance $C_p \approx 1.6\,\text{nF}$. In the head, $C_f = 47\,\text{pF}$ and $R_f = 1\,\text{M}\Omega$ are in parallel between the output and the summing node, and the + input sits at $V_{REF} \approx 1.45\,\text{V}$.

Feedback impedance:

$$Z_f = \frac{R_f}{1 + sR_fC_f}, \qquad Y_f = \frac{1}{Z_f} = \frac{1}{R_f} + sC_f.$$

Three equations describe the whole stage.

(1) Op-amp with finite gain (+ input at V_{REF}, so 0 in deviations):

$$v_o = A\,(0 - v_x) = -A\,v_x.$$

(2) KCL at the summing node. The piezo current leaves through C_p and through the feedback branch:

$$i_p = sC_p\,v_x + \frac{v_x - v_o}{Z_f}.$$

(3) The piezo:

$$i_p = \frac{dq}{dt} \;\longleftrightarrow\; i_p(s) = s\,Q(s).$$

We do *not* assume a virtual ground. v_x is allowed to be non-zero.

2 Exact solution

From (1), $v_x - v_o = (1 + A)v_x$. Putting this in (2):

$$i_p = v_x\Big[sC_p + (1 + A)Y_f\Big] \quad \Rightarrow \quad v_x = \frac{i_p}{sC_p + (1 + A)Y_f}.$$

Then $v_o = -A\, v_x$:

$$\boxed{v_o = -\, i_p\, Z_f\, \frac{1}{1 + \dfrac{1 + sC_pZ_f}{A}}}$$

Read this as **ideal result** $\times$ **error factor**. The ideal part $-i_pZ_f$ is what you get as $A \to \infty$ (the virtual ground). The numerator of the error term, $1 + sC_pZ_f$, is the *noise gain*.

3 Frequency response

A single-pole op-amp is modelled as

$$A(s) = \frac{\omega_t}{s}, \qquad \omega_t = 2\pi\,\mathrm{GBW}, \qquad \mathrm{GBW} = 50\,\mathrm{MHz}\ (\mathrm{OPA365}).$$

Substitute $i_p = sQ$ and $A = \omega_t/s$:

$$\frac{v_o}{Q} = -\,\frac{s\, Z_f}{1 + \dfrac{s\,(1 + sC_pZ_f)}{\omega_t}}.$$

Numerator. With $Z_f = R_f/(1 + sR_fC_f)$,

$$sZ_f = \frac{1}{C_f} \cdot \frac{s}{s + \omega_L}, \qquad \omega_L = \frac{1}{R_fC_f}.$$

Denominator. For $\omega \gg \omega_L$, $Z_f \approx 1/(sC_f)$, so $1 + sC_pZ_f \to 1 + C_p/C_f$. Define the noise gain

$$G_n = 1 + \frac{C_p}{C_f}, \qquad 1 + \frac{sG_n}{\omega_t} = 1 + \frac{s}{\omega_H}, \quad \omega_H = \frac{\omega_t}{G_n}.$$

Together:

$$\boxed{\frac{v_o}{Q} \approx -\frac{1}{C_f}\ \underbrace{\frac{s}{s + \omega_L}}_{\text{high-pass}}\ \underbrace{\frac{1}{1 + s/\omega_H}}_{\text{low-pass}}}$$

The head is a band-pass. Mid-band gain is $-1/C_f$.

Quantity	Formula	Value
Mid-band gain	$1/C_f$	21.3 mV per pC
Low corner	$f_L = \dfrac{1}{2\pi R_fC_f}$	3.39 kHz ($\tau = 47\,\mu$s)
Noise gain	$G_n = 1 + C_p/C_f$	$1 + 1600/47 = 35.0$
High corner	$f_H = \mathrm{GBW}/G_n$	1.43 MHz

At 250 kHz, the middle of the AE band:

$$\left|\tfrac{s}{s+\omega_L}\right| \approx 0.9999, \qquad \left|\tfrac{1}{1+j\,250/1430}\right| = \frac{1}{\sqrt{1+0.175^2}} = 0.985 \;\; (-0.13\,\text{dB}).$$

The perfboard's OPA1642 has GBW $\approx$ 11 MHz, so $f_H = 11/35 = 0.31$ MHz and the factor at 250 kHz is $1/\sqrt{1+0.8^2} = 0.78$, that is -2.1 dB. This is the 0.2 dB versus 2.2 dB difference between the boards.

4 How good is the "virtual ground"?

From the solution, $v_x = -v_o/A$. At 250 kHz, $A = \text{GBW}/f = 200$. For a 100 mV output, $v_x = 0.5$ mV. That is small but not zero. The charge that stays on C_p is

$$C_p v_x = (1.6\,\text{nF})(0.5\,\text{mV}) = 0.8\,\text{pC}.$$

This is the origin of the error term $G_n f/\text{GBW} = 0.175$.

> A large C_p compared with C_f means a large noise gain, less loop gain, and a larger error. A smaller C_f gives more volts per pC but more noise gain and less bandwidth. This is the design trade-off of the head.

5 Noise from the same model

Add the op-amp's input voltage noise e_n in series with the + input. With the signal off, the loop gives

$$v_{o,n} = e_n\Big(1 + \frac{Z_f}{Z_{C_p}}\Big) = e_n\,(1 + sC_pZ_f) \;\xrightarrow{\omega \gg \omega_L}\; e_n\,G_n.$$

This is the same quantity as in the error factor, which is why it is called the noise gain. With $e_n = 4.5\,\text{nV}/\sqrt{\text{Hz}}$:

$$e_n G_n = 158\,\text{nV}/\sqrt{\text{Hz}}, \qquad v_{n,\text{rms}} = 158\,\text{nV}/\sqrt{\text{Hz}} \cdot \sqrt{400\,\text{kHz}} \approx 100\,\mu\text{V}.$$

The noise densities are per $\sqrt{\text{Hz}}$ because independent sources add as *powers*.

Thermal noise of R_f appears as a current noise

$$i_{n,R_f} = \sqrt{\frac{4kT}{R_f}} = 0.129\,\text{pA}/\sqrt{\text{Hz}}.$$

It flows through Z_f. At 250 kHz, $|Z_f| \approx 1/(2\pi f C_f) = 13.5\,\text{k}\Omega$, so it adds about $1.7\,\text{nV}/\sqrt{\text{Hz}}$ at the output, about 1 % of the op-amp's. It can be ignored.

Equivalent noise charge (noise referred to the input):

$$Q_n = v_{n,\text{rms}}\,C_f \approx 100\,\mu\text{V} \times 47\,\text{pF} = 4.7\,\text{fC}.$$

The perfboard measurements show about 4 mV rms in the band, 40 times the amplifier's own noise. So the noise you see comes from elsewhere (plant, cables, pickup), not from the op-amp.

6 DC operating point

Set $s \to 0$ in the KCL. C_p and C_f are open circuits, so the only DC path is R_f. A real op-amp also has an input bias current I_b and an input offset voltage V_{os}:

$$V_{out,DC} = V_{REF}\frac{A}{1+A} + I_b R_f + V_{os} \;\approx\; V_{REF} + I_b R_f + V_{os}.$$

For DC the stage is a unity-gain follower.

- With I_b of a few pA and $R_f = 1\,\mathrm{M\Omega}$, $I_b R_f$ is of order µV. Negligible.
- If I_b were 160 nA (as in the THS4531), $I_b R_f$ would be 0.16 V. A high-bias part does not work with a megohm feedback resistor.

The reference comes from the divider R29, R30 (both 1 MΩ):

$$V_{REF} = V_+\frac{R_{30}}{R_{29}+R_{30}} = \frac{2.9\,\mathrm{V}}{2} = 1.45\,\mathrm{V}.$$

So the output rests at about 1.45 V, mid-supply. The available swing is about ±1.3 V:

$$Q_{max} \approx 1.3\,\mathrm{V} \times 47\,\mathrm{pF} \approx 61\,\mathrm{pC},$$

while events are 2 to 11 pC.

7 Supply down the coax (bias-T)

An inductor is the opposite of a capacitor: $|Z_L| = 2\pi f L$. For $L = 470\,\mu\mathrm{H}$:

$$|Z_L|(250\,\mathrm{kHz}) = 2\pi \cdot 250\,000 \cdot 470 \times 10^{-6} \approx 740\,\Omega \gg 50\,\Omega,$$

and ≈ 0 at DC. So the DC supply flows through the inductor, and the signal goes through the capacitor C38 instead ($|Z_C| = 1/(2\pi f C) = 6.4\,\Omega$ at 250 kHz for 100 nF). The drop across R26 (47 Ω) and the inductor's own resistance, at about 4.3 mA, takes roughly 0.4 V off 3.3 V, so V_+ at the amplifier is about 2.9 V.

8 Stability

The loop gain is $T(s) = A(s)\,\beta(s)$ with feedback factor $\beta = 1/G_n$. At high frequency $\beta \approx C_f/(C_f + C_p) = 1/35$, which is flat. The crossover ($|T| = 1$) is at $f_H = 1.43\,\mathrm{MHz}$. There the op-amp's single pole has used 90° of phase, leaving about 90° minus whatever the output stage (R_{31} and the load) adds. C_f in the feedback path is what keeps this stable: without it, C_p at the − node would add a feedback zero and could peak.

9 OPA365: the specs that matter

Spec	Value	Why it matters here
Gain-bandwidth	50 MHz	Sets the head bandwidth, GBW/G_n = 1.4 MHz. The perfboard's 11 MHz cost 2 dB.
Input voltage noise	4.5 nV/$\sqrt{\text{Hz}}$	Sets the noise floor: 100 µV rms at the output, 4.7 fC equivalent.
Supply range	2.2 to 5.5 V	The head runs at about 2.9 V, inside the range.
Rail-to-rail in/out	yes	The output can use the full swing around 1.45 V.
Zero-crossover input	yes	No offset jump when the input sits near mid-supply, where our reference is.
Quiescent current	about 4.6 mA	Almost all of the head's 4.3 mA. Reason for the OPA314 (150 µA) low-power option.
Input bias current	pA class (verify)	Allows R_f = 1 MΩ without a DC error.
Input offset voltage	a few hundred µV (verify)	Shifts the rest level slightly. Harmless with AC coupling.
Slew rate	about 25 V/µs (verify)	A 150 mV burst at 250 kHz needs $2\pi fV$ = 0.24 V/µs. Large margin.

The entries marked *verify* are quoted from memory. Check them against the datasheet.

10 The design in five steps

1. KCL at the summing node with $i_p = sQ$ gives $v_o = -Q\,sZ_f$ (ideal).
2. $Z_f = R_f \parallel C_f$: mid-band gain $-1/C_f$, high-pass at $1/(2\pi R_f C_f)$ = 3.4 kHz.
3. Finite $A = \omega_t/s$ adds the error factor with noise gain $G_n = 1 + C_p/C_f = 35$, which gives a low-pass at GBW/G_n = 1.4 MHz.
4. The same G_n multiplies e_n: about 100 µV rms at the output, 4.7 fC equivalent.
5. DC: a unity-gain follower to V_{REF}, plus $I_b R_f + V_{os}$.